

DevOps

Introduction to DevOps

DevOps is a modern software development and IT operations methodology that improves collaboration between development and operations teams.

Traditionally:

- Developers wrote code
- Operations teams deployed and maintained applications

This separation often caused:

- Delays
- Deployment failures
- Communication gaps
- Environment issues

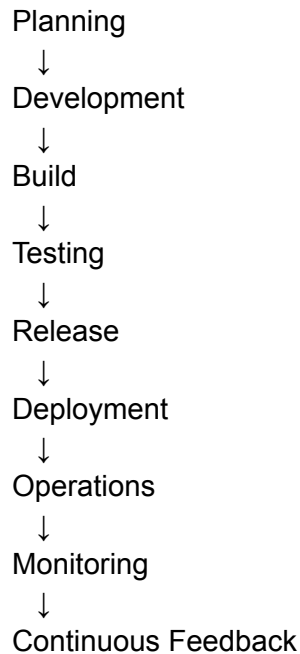
DevOps solves these problems by integrating development and operations into a continuous and collaborative workflow. The main objective of DevOps is:

“Build, test, deploy, and deliver software faster with better quality and reliability.”

Goals of DevOps

- Faster software delivery
 - Continuous integration and deployment
 - Better collaboration
 - High availability
 - Reduced failures
 - Faster recovery from issues
 - Automation of repetitive tasks
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DevOps Lifecycle



Phases of DevOps Lifecycle

Phase	Description
Planning	Requirements and project planning are done
Development	Developers write application code
Build	Source code is compiled into executable format
Testing	Automated/manual testing is performed
Release	Application is prepared for deployment
Deployment	Application is deployed to servers/cloud
Operations	Infrastructure and services are managed
Monitoring	Performance and errors are continuously monitored

Important DevOps Concepts

1. Continuous Integration (CI)

Developers regularly merge code into a shared repository.

Benefits:

- Early bug detection
- Better code quality
- Faster development

Common Tools:

- Jenkins
 - GitHub Actions
 - GitLab CI
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2. Continuous Delivery (CD)

Automatically prepares software for deployment after testing.

Benefits:

- Faster releases
 - Reduced deployment risk
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3. Continuous Deployment

Every successful change is automatically deployed to production.

4. Infrastructure as Code (IaC)

Infrastructure is managed using code instead of manual configuration.

Popular Tools:

- Terraform
- Ansible
- CloudFormation

Advantages:

- Consistency
 - Faster provisioning
 - Easy scalability
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5. Monitoring and Logging

Applications and infrastructure are continuously monitored.

Popular Tools:

- Prometheus
- Grafana
- ELK Stack

Benefits:

- Detect failures quickly
 - Improve performance
 - Better troubleshooting
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Common DevOps Tools

Category	Tools
Version Control	Git, GitHub, GitLab
CI/CD	Jenkins, GitHub Actions
Containerization	Docker
Container Orchestration	Kubernetes
Configuration Management	Ansible, Puppet

Monitoring	Grafana, Prometheus
Cloud Platforms	AWS, Azure, GCP

Advantages of DevOps

Advantage	Explanation
Faster Delivery	Software releases happen quickly
Better Collaboration	Teams work together effectively
Automation	Reduces manual tasks
Improved Quality	Continuous testing improves reliability
Reduced Failures	Automated pipelines reduce errors
Scalability	Infrastructure can scale easily
Faster Recovery	Quick issue detection and resolution
Cost Reduction	Efficient resource usage reduces cost
Security Integration	Security can be integrated into CI/CD pipelines

Challenges in DevOps

Challenge	Description
Cultural Change	Teams must adapt to collaboration
Learning Curve	Requires knowledge of multiple tools
Tool Integration	Integrating tools can be complex
Security Risks	Poor automation may introduce vulnerabilities
Initial Setup Cost	Infrastructure setup may require investment

What is Agile?

Agile is a software development methodology that focuses on:

- Flexibility
- Iterative development
- Customer collaboration
- Continuous improvement

Instead of developing the entire software at once, Agile divides the project into smaller parts called iterations. Each iteration delivers a working version of the software.

Agile Manifesto

The Agile Manifesto is based on four core values:

Agile Values
Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a fixed plan

Agile Principles

1. Early delivery of software
 2. Welcome changing requirements
 3. Frequent delivery
 4. Collaboration between business and developers
 5. Continuous improvement
 6. Simplicity
 7. Self-organizing teams
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Agile Workflow

Requirement Gathering



Planning



Design



Development



Testing



Feedback



Next Iteration

Advantages of Agile

Advantage	Description
Flexibility	Changes can be handled easily
Faster Delivery	Working software is delivered frequently
Better Quality	Continuous testing improves quality
Customer Satisfaction	Regular feedback improves product
Reduced Risk	Problems are identified early
Better Collaboration	Team communication improves
Continuous Improvement	Retrospectives improve performance

Disadvantages of Agile

Disadvantage	Description
Less Predictability	Scope changes frequently
Requires Active Participation	Customer involvement is necessary
Documentation May Reduce	Focus is more on working software

Difficult for Large Teams	Coordination becomes complex
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What is Scrum?

Scrum is one of the most popular Agile frameworks.

It helps teams manage and complete projects through iterative development.

Work is divided into small cycles called:

Sprint

A Sprint is a fixed-duration development cycle.

Usually:

- 2 weeks
- 3 weeks
- 4 weeks

At the end of each sprint:

- A working product increment is delivered
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Scrum Framework Components

Component	Description
Product Backlog	List of all requirements/features
Sprint Backlog	Tasks selected for current sprint
Sprint	Development iteration
Daily Scrum	Daily 15-minute meeting
Sprint Review	Demonstration of completed work

Sprint Retrospective	Discussion for improvements
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Scrum Roles

There are three main roles in Scrum:

1. Product Owner
 2. Scrum Master
 3. Development Team
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Product Owner

The Product Owner represents customer requirements and business goals.

The Product Owner decides:

- What to build
 - Which feature has higher priority
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Responsibilities of Product Owner

Responsibility	Description
Product Vision	Defines product direction
Backlog Management	Maintains product backlog
Prioritization	Prioritizes important tasks
Stakeholder Communication	Coordinates with clients/business
Requirement Clarification	Explains requirements to team

Skills of Product Owner

- Communication skills
 - Decision-making ability
 - Business analysis
 - Requirement management
 - Leadership
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Scrum Master

The Scrum Master ensures the Scrum process is properly followed.

The Scrum Master acts as:

- Facilitator
 - Coach
 - Problem solver
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Responsibilities of Scrum Master

Responsibility	Description
Conduct Scrum Meetings	Organizes daily scrum
Remove Blockers	Resolves obstacles
Team Guidance	Helps team follow Agile
Improve Productivity	Encourages efficient workflow
Facilitate Communication	Coordinates among teams

Skills of Scrum Master

- Leadership

- Communication
 - Conflict resolution
 - Team management
 - Agile knowledge
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Product Owner vs Scrum Master

Product Owner	Scrum Master
Focuses on product value	Focuses on Scrum process
Manages product backlog	Facilitates Agile practices
Works with stakeholders	Works with development team
Decides feature priorities	Removes blockers
Business-oriented role	Process-oriented role

Daily Scrum Meeting

A short meeting conducted daily for about 15 minutes.

Each team member answers:

1. What did I complete yesterday?
2. What will I do today?
3. What obstacles am I facing?

Purpose:

- Track progress
 - Improve communication
 - Identify blockers early
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Sprint Review

Conducted at the end of the sprint.

Purpose:

- Demonstrate completed work
 - Collect stakeholder feedback
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Sprint Retrospective

Conducted after sprint review.

Purpose:

- Identify improvements
 - Discuss team performance
 - Improve future sprints
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Agile vs Traditional Waterfall Model

Agile	Waterfall
Iterative development	Sequential development
Flexible requirements	Fixed requirements
Frequent delivery	Final delivery at end
Customer feedback throughout	Limited customer interaction
Faster issue detection	Issues found late

Agile vs DevOps

Agile	DevOps
Focuses on development process	Focuses on deployment and operations

Improves software development	Improves software delivery
Short development cycles	Continuous deployment
Customer collaboration	Automation and monitoring
Scrum/Kanban frameworks	CI/CD pipelines

Relationship Between Agile and DevOps

Agile and DevOps complement each other.

Agile

Focuses on:

- Planning
- Development
- Iterative delivery

DevOps

Focuses on:

- Deployment
- Automation
- Monitoring
- Maintenance

Together they enable:

- Faster releases
 - Better quality
 - Continuous delivery
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Product Owner – Requirements Document – Wireframes

- Product Owner gathers business requirements.
- Requirements document explains features and goals.

- Wireframes show basic UI design and user flow.

Technical Document – Architect & Tech Lead

- Architect and Tech Lead design the system architecture.
- Technical document includes tech stack, modules, APIs, and database design.

Coding Phase – SSE and SE

- SSE and SE develop application code.
- Follow coding standards and perform unit testing.

Testing Phase – QA Engineers

- QA team tests application quality and functionality.
- Includes bug testing, integration testing, and regression testing.

Deploy – Live – SSEs

- SSEs deploy application to production/live environment.
- Includes release management and monitoring.

Maintenance – SSE and SE

- Team fixes bugs and supports users after deployment.
- Handles updates, optimization, and improvements.

SDLC

- SDLC is the complete software development process.
- Covers planning, design, development, testing, deployment, and maintenance.